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FMS304 RESEARCH METHODOLOGY SUMMARY

MODULE 1

UNIT 1: CONCEPTUAL FRAME WORK OF RESEARCH

The Meaning of Research

Research may be broadly described as any systematic endeavours or striving towards the understanding, on perceiving certain complex situational problem of more than immediate personal concern and stated in a problematic form.

Travers (1969) has described research as an activity directed towards the development of an organized body of scientific knowledge about the events with which human beings are concerned. The goal is to discover general principles or interpretations of behaviour, to explain, predict and control events in everyday life situations.

Leedy (1997) defines research as the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon with which we are concerned or interested.

Leedy (1997) tells us that research is a process through which we attempt to achieve systematically, and with the support of data, the answer to a question, the resolution of a problem, or a greater understanding of a phenomenon. This process has eight distinct characteristics.

The Characteristics of Research

- Research originates with a question or problem: Everywhere around us is filled with many answered questions and unresolved problems. When we look around, we observe things that make us wonder and to ask questions. These questions may start to spark igniting chains of reactions which terminate in the research process. An inquisitive mind is the beginning of research.

- Research requires a clear articulation of a goal: It is critical to have a clear and unambiguous statement of the problem. This statement gives you an exercise in intellectual honesty. The ultimate goals of the research should be given in a grammatically complete sentence which is precise and clear.

- Research requires a specific plan of procedure: You should not hope that the data

necessary to solve the problem would somehow fortuitously come up. You have to have rather a planned attack, a search – and – discover mission explicitly planned or designed in advance.

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- Research usually divides the principal problem into more manageable subproblems: A whole is made up of parts. This is a natural law universally accepted. So when you think about your principal goal in research, try to observe this goal precept.

- Research is guided by specific research questions, problems or hypothesis: These guide you and direct you on what to do and how to do it in order to arrive at the solution to the problem.

- Research accepts certain critical assumptions: These must be self-evident truths. These assumptions must be valid in order to make the research to proceed. You must let others know what you assume with respect to your study.

- Research requires the collection and interpretation of data: You need to collect appropriate data, organize them in a meaningful way so that they can be analyzed and interpreted.

- Research is cyclical or helical: The process of research follows a cycle. It begins simply and follows logical developmental steps.

CHARACTERISTICS OF RESEARCH

Objectivity: In an ideal situation, a research is beyond the subjective bias of the researcher. As a researcher, you have to make deliberate efforts to eliminate all personal preferences. You should resist the temptation to seek only such data which support your hypotheses or your

Precision: When you conduct a good research and write the report but your readers do not understand what you have done, you may have succeeded in wasting your time and efforts. Every research should use a technical language in order to convey the exact meaning to the readers. Such research languages include validity, reliability, random sampling variables etc.

Design: If you want to have a good research, you must have a very good and systematic design. This implies that every scientific inquiry will generally undergo such steps as:

Definition of the problem

- Statement of the hypothesis
- Collection and analysis of data
- Testing and confirmation or rejection of hypothesis
- Reporting of the results

Verifiability: When you conduct a research, you write your report. This presents the research design and the findings to the professional community. From this point other researchers and scholars will study the report, analyze it in order to confirm or reject the outcomes. This tells you that research is a social enterprise. Its information is open for public scrutiny. Verification is related to objectivity and precision.

THE TYPES OF RESEARCH

Fundamental Research: The main purpose of these types of research is to obtain empirical data which can be used to formulate, expand or evaluate a theory. It is not actually directed in design or purpose towards the solution of practical problems. The main aim is to expand the frontiers of knowledge without the intention of having practical applications.

Applied Research: Unlike basic research, this type is directed towards the solution to an immediate, specific and practical problem. It is the type of research which you can conduct in relation to actual problems and under the conditions in which they are found in practice. You can use the applied research to solve problems at the appropriate level of complexity.

Quantitative and Qualitative Research: Quantitative research is an inquiry into a social or human problem, based on testing a theory composed of variables measured with numbers or figures and analyzed with statistical procedures in order to determine whether the predictive generalizations of the theory hold true. He also defines Qualitative research is an enquiry process of understanding a social or human problem, based on building a complex, holistic picture formed with words reporting detailed views of information, and conducted in a natural setting.

The qualitative research is sometimes referred to as the interpretative, the naturalistic, the constructivist or the postpositive approach.

THE PURPOSE OF RESEARCH

Training in Research

By going through this course, you are subjecting yourself to be trained on how to conduct researches. Training takes different shapes. It depends on whether the project you are undertaking is for the award of degree of Bachelors, Masters or Doctoral in research methodology.

Problem solving

This involves diagnosing and solving the problems in the system. You know your own system, banking and finance, tourism, Business management, hotel management among others. While we talk of problem solving, it is necessary you develop a more comprehensive view of it. It means a realistic understanding of a situation on the basis of data and statistics.

Search for Truth

This is the biggest challenge in research. A research in search of truth is most of the times classified as pure research. It is the ability to generalize and create knowledge. Such generalizations are derived from occurrences in repeated instances. Generalizations are drawn primarily on two bases. These are repeated observations in various possible situations and applications of statistical designs where the variables are statistically controlled.

DIFFERENCE BETWEEN QUANTITATIVE AND QUALITATIVE RESEARCH

Quantitative research is an inquiry into a social or human problem, based on testing a theory composed of variables measured with numbers or figures and analyzed with statistical procedures in order to determine whether the predictive generalizations of the theory hold true while Qualitative research is an enquiry process of understanding a social or human problem, based on building a complex, holistic picture formed with words reporting detailed views of information, and conducted in a natural setting.

UNIT 2.0: SKILLS OF SCIENTIFIC INVESTIGATION

THE SCIENTIFIC METHOD AND ACTIVITIES

The assumption of order: - This states that nature is ordered and not haphazard. All the events in nature follow each other in regular sequences. According to wood (1977) the assumption of order means that there is an overall pattern or scheme or order of events and it is discernible.

The assumption of determinism: - This states that events have courses, determinants or antecedents that can be detected. This means that events in nature are related and their patterns of relationships can be detected. That is to say that the relationship among events can be discovered.

The assumption of parsimony: - This state that the universe is organized and an explanation of natural phenomena should be simple and should be preferred to complex explanations.

The assumption of Empiricism: - this states that science investigations must be systematic and controllable. They should be empirical in using observable events in testing theories. The assumption as you can see refers to a reliance on field demonstration, observation and experiment.

FACTS AND THEORIES

Scientific enquiry starts with facts and then moves towards theorizing. To make the facts useful, they must be organized. So you now see that the primary purpose of the scientific method is to develop a mechanism for organizing the facts as they accumulate and become meaningful from the stand point of their objectives.

Scientists gather facts through empirical investigations. The accumulation of these facts brings about the need for investigation, organization and classification in order to make the isolated findings meaningful.

Theory is a set of interrelated constructs or concepts, definitions and propositions that present a systematic view of a phenomenon by specifying the relationship among variables, with the purpose of predicting and explaining the phenomenon. A theory knits together the results of observations in order to enable Scientifics make general statements about variables and relationship among variables.

We can then say that just as facts underlie theory, theories underlie facts. Both facts and theories raise themselves like a spiral to an increasingly precise scientific formulation. Theories have significance when they are built upon, classified and tested by facts. The

growth of science is dependent upon the accumulation of facts and the formulation of new or broader theories.

DIFFERENTIATE BETWEEN HYPOTHESIS AND THEORY

A hypothesis can be used to explain a small number of facts and the relationship among them. Generalization is a hypothesis based on broader phenomena. Theory is used to explain even more facts and their relationships. Theories themselves range from the simple to the more sophisticated. Finally, we have laws which have the greatest scope and generality.

LIST OF THE PURPOSE OF THEORY

- It summarized and puts the existing knowledge in a particular area in order.
- It provides a provisional explanation for observe events and relationships.
- It permits the prediction of the occurrence of phenomena and enables the investigator to postulate and eventually, to discover unknown phenomena

DESCRIPTION ON THE LEVELS OF THEORIZING

It is very important for you to know that good theories are not born out of imagination. They do not originate merely through arm chair reflection. A theory is built upon collected facts. You as an investigator do some searches, make intelligent guesses as to how the fact are ordered, you are missing ideas or link and put forward a hypothesis. You then deduce facts which are consistent or otherwise with the deductions.

You build a wider generalization or conceptual framework on more facts. Eventually, you outline a theory. Theories are solidly based on evidence. They are very important practical tools which enable us to advance our knowledge still faster.

THE LEVELS OF THEORIZING IN RESEARCH

Psychologists agree that it is better to keep theorizing down to the levels that involve a minimal use of abstract terms. You have to remember this when you do some theorizing. There are six levels of theorizing.

Level 1: Hypothesis formation: - if you are going to use any hypothesis as a basis for your research, such hypothesis must go beyond the fact on which it is based. You need to establish the existing state of affair. Most of the surveys are conducted for this purpose. You may conduct a survey of what customers of a hotel think of a particular aspect of the services provided for them, how much time is spent by your customers in watching television etc

Level 2. Elementism: - Primitive forms of classification in terms of some significant set of ideas are example of theories at this level. In the case of hotel management, such classification of supervisory abilities of hotel stuff or accounting abilities derived from factor

analysis or the classification of hotel acts that may be the result of extensive observation in hotel management would come under the level of elementism.

Level 3. Descriptive theories and taxonomies: - when you formulate descriptive theories and taxonomies, you are operating at a more complex level than mere classifications. A hierarchal categorization of different cases varying from the most simple to the most complex, example is the Bloom's taxonomy of cognitive behaviour.

Level 4. Classification: If you want to develop any kind of useful classification system, there must be a set of theoretical ideas to underline the classification. For instance, the classification of animals became significant only after it was realized that the species could be fitted into a system in which the evolutionary relationships among animals became the basis for classification.

Level 5 & 6. Postulates and Theories: The higher levels: - here we have the highest levels of theorizing. They can only be observed in the physical sciences. Here, a theory consists of a set of basic statements called axioms of theory, which tell us what the theory is all about. This axiomatic statements or postulates include primitive terms which are initiatively understood and cannot be otherwise completely defined

UNIT 3.0 METHODS OF DATA COLLECTION IN RESEARCH – 1

DATA COLLECTION IN RESEARCH

Data can be regarded as information. It can be given out or taken for the purpose of making interferences in research. In this case, it may include numerical or statistical results or figures such as percentages. It may also include verbal materials like newspaper accounts, scholastic essay etc. we can therefore say that any collection of verbal or numerical information from which inferences or conclusions can be drawn or analyzed is regarded as data. It means that data can be quantitative or qualitative.

Any information which comes in numbers, figures, measures or quantities is said to be quantitative. Where as any information which comes as a verbal description of attribute or characteristics is regarded as qualitative. According to Tolbert (1967) all data gathering devices should be closely related to the design of the study.

THE TYPES OF DATA USED IN RESEARCH INVESTIGATIONS

There are basically two categories data collected and used in research. These are: - Primary data and Secondary data

1. **Primary data**

These are information, facts or statistical materials which you as a researcher originate for the purpose of the inquiry on hand. This is sometimes popularly called the "First hand information" or "information from the horse's mouth". They are referred to as eye witness account of an event or phenomenon. Such information is extracted from the actual participants themselves. It may be through oral interviews or discussions or through written diaries, minutes, proceedings, pictures, objects etc.

2. **Secondary data**

These refer to information, facts or statistical materials which are not originated by you as the researcher or investigator. These are materials from someone else's records or other documents like books, journals, newspaper reports and other research works that may be got from the libraries. In the simplest form, secondary data are not first hand information.

QUESTIONNAIRE METHOD OF COLLECTING DATA

A questionnaire is used when factual information is desired (Best and Khan 1995). When opinions rather than facts are desired; an opinionates or attitude scale is used. A questionnaire is a self reporting instrument that has received a good use in educational, researches, psychological and social science researches, programme evaluation etc. it is described as the most common type of research instrument.

THE CHARACTERISTICS OF A GOOD QUESTIONNAIRE

- A questionnaire deals with a significant topic.
- It seeks only such information which cannot be obtained from other sources
- It is as short as possible, and only long enough to get the essential data
- It is attractive in appearance
- Instructions are clear and complete

CONSTRUCTION OF A QUESTIONNAIRE

Henderson et al (1978) suggested eight steps to which the process of developing and using a questionnaire can be divided. These are;

- Identifying the programmes objectives and specific information to be obtained
- Select a response format
- Identifying the frame of reference of the respondents
- Writing the items/questions
- Preparing a data summary sheet
- Critiquing the questions, trying them out and revising them
- Assembling the questionnaires
- Administering the questionnaires

TYPES OF QUESTIONNAIRES

There are two main types of questionnaires based on two basic types of question formats. These are closed ended questions and open-ended questions.

Closed ended questionnaire are those that provide respondent with fixed set of alternatives from which they are to choose where as open-ended questionnaire are those questions to which the respondents write their own response, as it is in an essay examination questions.

The Closed – Ended Questionnaire

It can also be called restricted or closed form type. All questionnaire instruments that call for short, check-mark responses belong to this category. For example, you may be asked to mark yes or no, check an item from a list of suggested responses

The Open – Ended Questionnaire

This can also be called unrestricted questionnaire or free response type. It demands that the subjects respond in their own words. Look at this question again. Why did you choose to study Hotel Management? You can see that no clue is given here. The advantage is that open form provides for greater depth of response. The respondents may reveal their frame of reference and sometimes the reasons for their responses. It has such disadvantages as: very low returns due to greater efforts needed on the part of the respondents in filling the questionnaire.

Pilot testing of the Questionnaire

Every instrument used for data collection in research should be trial tested or pilot tested. There are so many reasons for doing this. Among these reasons are:

- To find out how long it takes the respondents to complete
- To ensure that all questions and instructions are clear
- To enable the researcher to remove any items which do not yield usable data
- To enable you include any item deemed necessary to yield some relevant data.
- To use the results to find the reliability of the instrument.
- To use the results to do the item analysis of the instrument
- To ensure the usability of the instrument

Administering the Questionnaire proper

- Administering the questionnaires personally
- Using research assistants
- Mailing the questionnaires

Covering letter

Remember to accompany your questionnaire with an introduction letter or a covering letter. This should inform the respondents about the objectives of the study and therefore the questionnaire. You need to let them know that will be done with the information collected. They need to know the deadline for the return of the questionnaire. It will also assure them about the anonymity and confidentiality of the information supplied.

UNIT 4.0 INTERVIEWS AND OBSERVATIONS

WHAT IS AN INTERVIEW

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An interview is a face to face oral interaction between the investigator and the respondents. You as the investigator, otherwise called interviewer or your assistant as the case may be read the questions directly to the respondents. The responses are recorded for analysis. The interview gives you a degree of flexibility which is not available in the questionnaire. The research interview is not as widely used as the questionnaire. The reason is that it takes a great deal of time and fund to conduct.

The interview technique is used routinely in normative survey research. During the process, the respondents or interviewees respond to some sort of direct questions posted to them by the investigator. The major purpose of interviewing is to find out what is in the mind of the interviewee.

THE TYPES OF INTERVIEW

The different types of interviews used in research are:

Informal Conversation Interview: This is characterized by the fact that the questions emerge from the immediate context or situation and are asked in the natural course of things. There are predetermined questions or topics or wordings.

The advantages Informal Conversation Interview include

- It increases the silence and relevance of the questions
- Interviews are built on and emerge from observations
- The interview can be matched to individuals and circumstances

The disadvantages Informal Conversation Interview

- Different information's are collected from different questions
- It is less systematic and comprehensive
- Data organization and analysis can be quite difficult

Interviewer Guided Approach: In this type, topics and issues to be covered are specified in advance, in an outline format. You as an interviewer will decide the sequence and wording of the questions in the course of the interview.

The advantages include:

- The outline increases the comprehensiveness of the data
- It makes data collection somewhat systematic
- Logical gaps in data can be anticipated and closed
- Interview is fairly conversational and situational

The disadvantages include:

- Important and silent topics may be inadvertently omitted.

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- Interviewer flexibility in sequencing and wording of questions can result in substantially different responses from different perspectives.
- This will reduce the comparability of the responses.

Standardized Open-ended interview: In this case, the exact wording and sequencing of the questions are determined in advance by the researcher. All interviewees are asked the same basic question in the same order. Questions are worded in a completely open – ended format.

The advantages include:

- All respondents answer the same questions. This increases the comparability of the responses.
- Information collected are complete for each respondents on the topics addressed in the interview.
- It reduced interviewer effects and biases when several interviewers are used.
- It permits the users of evaluations to see and review the instrument used for evaluation.
- It facilitates organization and analysis of the data.

The disadvantages include:

- Little flexibility in relating the interview to particular individuals and circumstances.
- Limited and constrained naturalness and relevance of questions and answers.

Closed or Fixed Response Interview: In this type which is the most popular in research studies, questions and response categories are determined in advance. Responses are fixed. The respondents choose from among the fixed responses provided.

The advantages include:

- Data analysis is simple
- Responses can be directly compared and easily aggregated
- Many questions can be asked and answered in a short time

The disadvantages may include:

- Respondents must fit their experiences, knowledge and feeling into the researcher's response categories
- Sometimes it may be seen as impersonal, irrelevant and mechanistic
- There can be a distortion of what respondents really mean or experienced by so completely limiting their response choices.

THE STEPS IN THE ARRANGEMENT OF INTERVIEWS

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For the successful handling of the interviews as a data collection technique for your research studies, you will need to follow some simple but very important steps.

These steps include:

- Set up the interview plan used in advance.
- Prepare and send the set of questions which you will ask to the interviewee in advance
- Sort for permission to tape-recorded the interview
- Confirm the date of interview in writing
- Send a reminder, together with the same set of questions to the interviewee, about 10 days to the interview date.
- On the interview day, be prompt, have another copy of the set of questions for your interviewee, should he/she has misplace his/her copy?
- Follow the questions serially as it is in the set.
- After the interview, prepare and submit a type script of the interview either in soft or hard copy to the interviewee tries to get a written acknowledgement of its accuracy or a corrected copy from the interviewee.
- Ask for final approval and written permission to use the information in your research report.
- After you have incorporated the information in your research report, send the relevant section of the report to the interviewee.

HOW TO CONDUCT AN INTERVIEW

An interview is a social interaction or relationship designed to exchange information between the interviewer and the interviewee. The goal is to collect data and not make friends. Therefore the quantity of information exchanged or received depends on how astute and creative you are at understanding and managing the relationship.

In order to obtain valid interview results Schofield (1972) advised the researcher to:

- Listen to the respondent in a friendly and patient but intelligent critical manner.
- Refrain from showing an authoritarian attitude Refrain from giving advice on moral instruction.
- Avoid asking questions which compel the subject to give the reply which you want to have.
- Avoid leading or prompting the subject towards certain answers.
- Avoid entering into argument with your interviewee.
- Avoid the use of double-barreled questions.

Strengths of the interview

The interview is often regarded as being superior to other data-collecting devices. One of the reasons is that people prefer to talk than to write, especially when it comes to controversial issues. When the interview has gained support or when you have established a friendly relationship with your interviewee, certain confidential information may be obtained. Such

information that an individual might be reluctant to put in writing could be elicited from subjects.

Recording the Responses

As an interviewer, your central task is to record the responses from the respondents. There are four most common ways of recording responses. These are:

Classifying responses into predetermined categories; summarizing the high points of what is said; taking verbatim notes or recording the interview with a tape recorder or video machine.

To record the responses you can be taking notes as the interviews are going o, or take the notes immediately after, when the information is still fresh in your mind. This is called mental note. Alternatively, you can record the interview with recording machines.

OBSERVATION AS A DATA COLLECTION TECHNIQUE

Observation

Observation can be described as the act of looking out for and recording the presence or absence of both verbal and non-verbal behaviour of an individual or group of individuals. According to Monette, et al (1994) observational technique is the collection of data through direct visual or auditory behaviour or experience. It includes video or audio recording of behaviour.

When the observation of natural phenomena is aided by systematic classification and measurement, it leads to the development of theories and laws. Every type of research, whether experimental, descriptive or qualitative, incorporates the use of observational techniques. It means that you as a researcher looks or listens in order to see or hear the behaviours or words respectively. These provide the information needed for the research.

There are two main types of observation:

Participant observation

In this case the researcher become part and participates in the activities of the people, group or situation that is being studied. Participant observation is very good for a highly qualitative research.

The non-participant observation

In this case the researcher is not part of and does not in any way participate in the activities of the people, group or situation. He observes through recording instruments or from outside.

THE CHARACTERISTICS OF A GOOD OBSERVATION

According to Best and Khan (1995) the following standards should be the characteristics of observers and their observations. These characteristics are:

Observation is carefully planned: Observers who are systematic and perceptively know what to look for. They are not distracted by irrelevant situations.

Observers are aware of the wholeness of what is observed: They are alert to significant details. They know that the whole is often greater than the sum of its parts.

Observers are objective: They recognize likely bases and strive to eliminate their influence on what they see and report.

Observers separate the facts from the interpretation of facts: They observe the facts and make their interpretations later.

Observers are objective: They recognize likely bases and strive to eliminate their influence on what they see and report.

Observations are expertly recorded: They use appropriate instruments to systematize, quantify and preserve the results of their observations.

Observations are made in such a way as to make sure that the data collected are valid and reliable.

THE USES OF OBSERVATION

- Measuring classroom process variables
- Measuring attainment of programme objectives
- Measuring the extent of programme implementation
- Identifying difficulties in programme use
- Identifying changes introduced by hotel owners, teachers etc
- Identifying typical pathways to customer services
- Supplementing data from other sources

THE THINGS TO BE OBSERVED

There are some general categories of things you have to observe and record. These include

The Setting: This could be a hotel, a bank, a company etc. The records should contain descriptions of the general physical and social setting being observed.

The People: These include customers, staff, management etc. The records should include a physical and social description of the main characters who are the focus of your observation

Individual Behaviour: The cares of observations in most studies are the behaviour of the individuals in the settings.

Group Behaviours: Most of the times, the behaviour of a group of individuals like customers, staff etc. may provide an important bit of information for your studies

THE RECORDING OBSERVATION

There are different kinds of devices used extensively to aid the recording of information gained through observation. These devices include checklist, rating scales, scorecards, scaled

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specimens etc. these provide systematic means for summarizing and/or quantifying data collected through observation or examination.

For instance, you can prepare a type of form or sheet of paper to rate such things as type of food served, quality of served, neatness of the staff, neatness of the rooms, neatness of the bidding, services provided, entertainment provided etc. such assessment sheet from the observational scale which you can use for your studies.

UNIT 5.0 PROBLEM IDENTIFICATION IN RESEARCH

THE CONCEPT OF PROBLEMS IN RESEARCH

In a research process, you have to select a topic of study from your area of study. After this you narrow down this topic for your study. The next concern is to define the problem which you are focusing on. Kerlinger (1979) defines a research problem in two ways.

The first is in a general sense in which he regards a research problem as a question that states situation needing discussion, inquiring, decision or solution.

The second one which is a more satisfactory definition, he regards a research problem as a question that asks for how two or more variables are related.

THE CHARACTERISTICS OF RESEARCH PROBLEM

According to Tuckman (1972) some characteristics of a problem in research are outlined as follows:

- It should ask about a relationship between two or more variables
- It should be stated clearly and unambiguously, usually in question form
- It should be possible to collect data on it so as to answer the question(s) asked
- It should not represent a moral or ethnic position indicating bias or feeling of preference

From these you must note that every research topic must have two more variables that can be measured or manipulated or controlled. The research problem must ask for the relationships between the variables contained in it. E.g. what is the relationship between the socio-economic status of parents and the intelligent quotient of their children? A problem must be researchable i.e. testable through empirical methods of field or laboratory observation and collection of data.

Remember that your research problem should be an empirically testable proposition stated in a question form containing two or more measurable or manipulative variables whose relationships are sort for in the problem statement itself. As a researcher, you are required to state your research problems in accordance with some principles.

These principles listed by Egon Guba (1973) are as follows:

- State the problem in terms intelligible to someone who is generally sophisticated but who is relatively uninformed in the area of the problem
- Define and delimit the specific area of the research
- Fore-shadow the hypothesis to be tested or the questions to be raised in the problem statement.

SOME SOURCES OF SUITABLE RESEARCH PROBLEMS

Most of the problems confronted in the school system, the community, banking industry, hotel and hospitality industries, etc, lend themselves to investigations.

The whole area of computer education and literacy provide avenue for a good number of research topics through the internet and websites. You can also get topics through international abstracts typed for computer processed.

Classroom lectures, class discussions, seminar reports and out of class exchanges of ideas with fellow students and lecturers can suggest many stimulating problems for solution. When you read assignments, journal articles, textbooks, research report, term paper, etc can also suggest additional area of needed research.

Some research topics can be selected on the basis of their use in verifying and testing a particular theory. You know that every research has some implications for existing theory. Therefore some research topics are selected specially for the purpose of testing some aspects of a given theory. In your exploration in an area, you may come across additional problems that need resolving. Most of these problems are often theoretically framed.

Another fruitful source of research problem is prior research. All research projects have limitations, new questions may be raised by the findings, research reports have discussions of weakness and limitations of the research, including suggestions for further research, so focusing on these unanswered questions or expending on previous research is a good way to find research problems.

Programme evaluation and practice effectiveness evaluation can form important activities for research purposes. Other sources of problem selection include political issues.

SELECTING A RESEARCH PROBLEM

Selecting a research problem is one of the most difficult tasks you have to face in the research process. This statement of fact is not in anyway intended to discourage you. According to Best and Khan (1995) one of the most difficult phases of the graduate research project is the choice of a suitable problem. This is because most students often select problems that are too broad in scope.

According to Good et al (1945) you should painstakingly select your problem of study upon these considerations:

- Novelty (newness) of the problem
- Your interest in the problem
- Practical value of the research on the problem to you and others in the society

- Your special qualification to handle the problem
- Availability of data on the problem
- The cost of doing the investigation on the problem
- The time required and available for the completion of the study

The sources of research problems include:

- Problems from the schools, community, banking, hotel etc.
- Technological changes and developments
- Computer education or literacy
- Classroom lectures and discussion
- Seminar reports Exchange of ideas
- Lectures Assignments Journal articles
- Textbooks Research reports
- Term paper
- Verifying & testing
- Prior research
- Political issues
- Programme evaluation
- External interests Supervisors

UNIT 6.0 STEPS IN RESEARCH AGENDA

THE COMPONENT OF SETTING THE PROBLEM

In every research process or encounter, there are components which help in shaping and refining the problem in order to provide a focus and manageable scope for the research. These components help you to quicken the pace of the study and that of the reader of the research report in understanding it. Before we look at these components, let us first list the steps which you will follow when conducting your research.

Step I. identifying the problem

You have already studied this step in details in the last unit. But let us say that it involves the discovery and definition of scientific research problem within a topic area which you have chosen to study. It may include questions which require answer in our area of study. For instance, what are the causes and effects of lecturers‟ strike on the standard of Education in Nigeria? Are men better managers than women?

Step II. Constructing Hypotheses

When you identify a researchable problem, it automatically leads to the proposition of a hypothesis as a tentative answer. Once you have identified a problem, you use the logical processes of deduction and induction to formulate an expectation of the outcome of the study. This expectation should be stated and tested. This implies that you are hypothesizing about the probable relationship existing between the concepts, variables identified in the problem. The hypotheses are the pointers to your solutions.

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Step III. Identifying and labelling the variables

The next step is to identify and label all the variables in the problem and hypotheses stated. There are different types of variables in research.

Step IV. Proposing Operational definition of terms

All terms, concepts and variables should be converted from their abstract or conceptual forms to operational forms that can be quantified or measured using some forms of specified measurement instruments. In technical terms, when you aperationalise the variables in your research, it means that you are stating them in observable and measurable forms which make them available for clear understanding, definition, manipulation, control and examination.

Step V. Manipulating and controlling variables

If you have to manipulate and control the variables in your study, you must understand the concepts of internal and external validity. This is because you need to control or manipulate and regulate these variables in order to study the relationship between variables in research.

Step VI. Constructing the Research Design

You can describe a research design as a specification or plan of operation for the testing of a stated hypothesis under a given set of conditions. It comes in specific types such as one-variable design, factorial design, experimental and quasi-experimental design etc.

Step VII. Identifying and constructing data collection devices

After defining your variables and choosing the design for the study you need to adopt, adapt or develop some instruments for measuring the selected variables.

Step VIII. Constructing Questionnaires and Interview Schedules

Most research studies in the social sciences and business rely on questionnaires and interviews as their source of data. You should employ recommended guidelines to develop and use these measurement devices for your data collection to test your hypotheses.

Step X. Carrying out Statistical Analysis or Data Analysis

The data you have collected above must be reduced, arranged and presented in an arranged and presented in an organized form for easy analysis. The data as organized are subjected to specified or suitable statistical indices for testing the hypothesis in order to generate some results or findings. From these conclusions, generalizations are drawn. You can employ the use of computers for easy and accurate data analysis. This aspect is very sensitive and requires some professional expertise, care and endurance.

Step XI. Discussion of Research Findings

You have to discuss at this stage, your research findings or the results from the data analysis. This is to justify, interprete, explain and further develop a given theory for knowledge based on the findings generated. This is where you show your disposition and prowess.

Step XII. Drawing Conclusions, Generalizations and Recommendation

At this stage you are expected to draw your conclusions from your findings and to make generalizations of the findings generated from the samples to the larger population from

which the sample was drawn. It is very important that your findings must be valid, your generalizations must be correct and your recommendations from the research findings must be possible for application so as to bring progress in the development and practice of the area.

Step XIII. Writing the Research Report

Writing the final research report is very important and constitutes the major secretariat or clerical activities of the research process. There are different styles and formats with their respective instructions and guidelines. You can therefore choose a desired format in order to produce your final research report.

The American Psychological Association (APA) format is the most popular and most used in research. Your supervisor will make it available to you if you demand it.

Delimitations

Delimitations give you the boundaries of the study. The conclusions are not to be extended beyond these boundaries. It tells you about the target population, what the research intends to do and what he intends not to do. When stating the delimitations, you are required to rule out all variables and issues considered irrelevant to the research. It is not the same thing as limitations.

For the purposes of clarification limitation are those conditions beyond the control of the researcher that may place restrictions on the conclusions of the study and their applications to other situations. These may include the use of unvalidated instrument, inability to randomly solid and assign subjects to experimental and control groups, some administrative policies etc.

Assumptions

These are statement of what the researcher believes to be facts but cannot be verified. They are what the research takes for granted. You have to state your assumptions in a study because it is easier to evaluate your conclusions based on the assumptions. You should not leave anything on your problem must be clearly and unreservedly spelt out.

The Significance of the Study

What is the use of the study? What are the practical values of the study? Who are the beneficiaries of the findings and how would they benefit and how the study will be useful to them?

Definition of Terms

You have to define all unusual terms that could be misinterpreted. These definitions will help you to establish the frame of reference with which you approach the problem. The definitions are in operational forms. Without explicitly knowing what a term means, you cannot evaluate the research or determine, if you have carried out what you set out as the main objectives of your study. The definitions must interpret the terms as they are used in relation to the study.

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This means that you determine what you wish the term to mean within the contact of the problem or its sub-problem.

HYPOTHESIS

Hypothesis is a tentative proposition set forth a possible explanation for an occurrence or a provisional conjecture to assist in guiding the investigation of a problem. Hypotheses and research questions are very helpful in research because you need some points around which to orient the research in searching for relevant data and in establishing tentative goals against which to project the data.

Hypotheses can be stated in two different forms. These are the null hypothesis, H0 and the alternative hypothesis H1.

The Null Hypothesis H0

This type relates to a statistical method of interpreting conclusions about population characteristics that are inferred from the variable relationships observed in samples. It asserts that observed differences or relationship merely results from chance errors inherent in the sampling process. It is a non-directional hypothesis. e.g.

- There is no significant difference in the academic performance of day students and boarders.
- There is no significant difference in the spending pattern of men and women in relation to fashion.
- There is no significant difference in the choice of accommodation between the young and elderly people.

Alternative Hypothesis H1

This is directional hypothesis. It gives the directory of the relationship between the variables. E.g. Women spend more money in fashion than men. Young people choose costlier accommodations than the elderly people etc.

VARIABLES

A variable is a property that takes on different values. According to Kerlinger (1977) a variable is a symbol to which we assign numerical values.

Some important variables in the humanities are sex, income, social class, aptitude, anxiety etc. A variable may have only two values e.g. male, female, dead or alive etc. Some variables used in behavioural researches are true dichotomies. This means that they are characterized by the presence or absence of a property e.g. good-bad, employed-unemployed etc. Some variables are polytomies e.g. the Nigerian citizens can be Igbo, Hausa, Yoruba, Efik, Nupe, Gwari, TIV, etc.

Most variables are theoretically capable of taking on continuous values e.g. intelligence, achievement etc.

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The different types of variables are:

1. **Independent and Dependent Variables**

An independent variable is the presumed cause of the dependent variable which is the presumed effect. The independent variable is the antecedent while the dependent variable is the one that is consequent. In experiments, the independent variable is the one that is manipulated while the dependent variable is the effect of the manipulation.

2. **Discrete and continuous variables**

Discrete variables are those with finite number of distinct and separate values. They have only whole numbers and no fraction e.g. sex, race, number of days, family size etc. Continuous variables are those that least theoretically can take an infinite number of values. It can take both number and fractions e.g. age, test scores etc. there are other types of variables.

Feasibility of a Research Problem

This implies that before you have a full scale investigation into the problem of your study, you would have considered in clear terms the practical issues involving feasibility. It involves consideration of practical issues of what can be reasonably accomplished given the time and resources available. This will help you to decide to reduce the scope or to increase it.

The major aspects to be considered are:

1. **Time constraint**

- Population
- Proper development of measuring devices
- Time required for data collection.

2. **Financial Constraint**

All the financial expenses for the study should be assessed. The major areas of financial involvement include: Production of instrument, payment for research assistants, transportation cost, cost of analysis of data, office supply and equipment and miscellaneous.

3. **Anticipating and Avoiding Problems**

Whether you are knowledgeable in research or not, you have to be able to identify potential trouble spots in your proposal. You can make some modifications to avoid them. You need to conduct a pilot study. This involves a preliminary run through on a small scale of all the procedures that will be employed in the study. If there are problems in the pilot study they can be estimated and taken care of. If the problem is that of time or money, you can scale the problem down by reducing the sample size, the number of hypothesis etc.

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UNIT 7.0 THE TYPOLOGIES OF THE RESEARCH METHODS I

EMPIRICISM

Any system which does not agree with all past knowledge which rests solely on experience can be regarded as empiricism. In other words it lays emphasis on induction, the use of direct observation to confirm ideas and the linking together of observed facts to form theories or explanations of how natural phenomena and events work. An empiricist is one who practices empiricist.

Empirical Enquiry

An empirical scientific research can be regarded as a way of looking at things. The facts of science are the products of observations. The construction of a theory is the most important purpose of empirical scientific inquires. Theory gives orientation to empirical inquiries it guides the collection of relevant date and offers conceptual scheme that will bind them together in a more systematic manner, theories are said to represent improvement in knowledge.

Facts clarify and redefine theory. But theories and facts stimulate each other and contribute to the growth of empirical knowledge.

Empirical studies make use of all research tools. These include observations, both participant and non-participant; interviews which can be structured, semi-structured and in-depth; key informant testimonies, analysis of personal and institutional documents, mass media analysis, examination of official documents and statistics, archival searching, review of published literature.

Empirical social research employs a wide range and variety of analytical techniques such as ethnographic interpretations, historical reconstructions, action research, multivariate analysis structuralist deconstruction and semiological analysis.

Empirical Research Process

In empirical research method, you proceed in a systematic and orderly manner in order to pursue the truth as determined by facts and logical considerations. The main purpose is to find a systematic interrelation of facts by experimentation, observation and logical procedures.

The steps involved are:

- Data collection through careful and critical observation with patience, precision and impartiality
- Measurement
- Classification, Organization and Tabulation of data
- Data analysis and Reduction
- Formulation of hypotheses
- Formulation of theory and Law

The main features of empirical scientific research are:

1. **Observation**: In order to obtain knowledge, empirical research relies onobservation. It is the cornerstone of any empirical inquiry. It supplies us with the thing that we try to understand and explain observation in an empirical research is not casual, but a conscious and deliberate activity designed to reduce error. It requires improved accuracy. It uses certain devices which add great precision to scientific observation. Meaningful observation results in the establishment of facts. These are used for verifying empirical theories, building them and modifying or improving them.
2. **Concepts**: These are the building blocks of scientific empirical research. Theories consist of several inter-related concepts. Although concepts are said to be fundamental to all human communication and thought, yet the concept we use in our day to day affairs are not always clear and precise.
3. **Objectivity**: This indicates that all the conclusions you reach in your empirical research are not affected by your personal views, values or biases. It permits repetition of observations under practically identical conditions, to get the same results. It facilitates the verification of facts and theories by many empiricists independently.
4. **Verifiability**: The findings of your research are said to be empirical if only they can be verified. Any theory which contains facts that cannot be verified tends to be metaphysical rather than scientific. Therefore all empirical conclusionsare liable for verification at any time. An empirical scientific theory of generalization stands to be rejected or modified at anytime.
5. **Predictability**: Accurate and precise predictions constitute one of the most impressive achievements of empirical researches. Predictability depends on the nature of the phenomena and our knowledge of the causes of the phenomena. Prediction also depends on our ability to have precise theories and accurate measurement.
6. **Systematic Nature**: Empirical investigations are systematic, thorough and rigorous in making use of designs to guard against errors from data collection, interpretation and generalization from the data. Empirical investigations are systematic in the methods used for data collection, formulation of hypotheses, data analysis techniques and the logic use for making generalizations.

PHENOMENOLOGY

This is a philosophy of knowledge which emphasis directs observation of phenomena. It refers to a person‟s construction of the meaning of a phenomenon as opposed to the phenomenon as it exists external to the person. It is a research method that attempts to understand the participant‟s perspective and views of social realistic. The phenomenologists seek to perceive reality and to describe it in words rather than numbers. It is a qualitative research employing the use of words that reflect consciousness and perception.

The phenomenologist concentrates on the phenomena and tries to produce convincing descriptions of what they experience rather than ordinary explanations and causes. The phenomenon experienced or studied may be an event, a relationship, an emotion or a programme.

Phenomenologist try to understand that a particular experience is all about by describing it as it is found in concrete situations and as it appears to the people who are involved in it. The central qualities of phenomenological research are attention to experience and intention to describe the experience.

Phenomenological Approaches in Social Research

Hermeneutical Phenomenology: This is one of most influential phenomenological field work. It involves a dialogue between a text, such as myth, drama, fairy story, dream report, oral history, etc and the experiences evoked in people participating in the text. The meaning of the text can be developed within the consciousness of living people

Transpersonal Phenomenology: This involves the recognition of extraordinary experiences as legitimate and useful data. Such experiences that in some sense go beyond the boundaries of ordinary ego-consciousness are called extraordinary. They include such phenomena as out-of–body experiences, visions, possession states, near-death experiences, meditative, ecstatic, unique and mystical experiences.

Social Phenomenology: This has become increasingly an influence on anthropological thinking of the social dimensions of experience. The object of scrutiny is your relationship to another person. It is not about the non-human objects of the world.

Neuro-Phenomenology: This provides the most direct route to uncover the essential structures of consciousness. You can do this by stepping in the cross-cultural evidence that pertains to human experience and to explore the universal structures of experience. The neurosciences provide an independent source of looking directly at the architecture of the organ of experience – the human brain.

CRITICAL RESEARCH

Critical research lays emphasis on the fact that knowledge is problematic and capable of systematic distortion. Its concern is to understand the theory as well as the practices. It is extremely varied the critical methodology based on a number of building blocks. These blocks should not be considered as discrete units which can simply be placed next to one another. They are elements which are drawn together in various ways in the process of deconstruction and reconstruction.

The Elements of Critical Social Research

The elements of Critical Social Research are

- Abstraction
- Totality
- Essence
- Praxis
- Ideology
- History
- Structure

Abstraction

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This is often misunderstood in term of a distillation of sensory perception of the world of objects into conceptual categories. It starts from the literally objective world and selects out the recurrent or apparently the core or the defining features until an abstract concept is formed, at least in our minds if not in a directly communicable form. It works by moving from the abstract to the concrete. It starts with abstract generalizations and then to their investigations.

Totality

Totality refers to the view that social phenomena are interrelated and form a total whole. It means that a social phenomenon should be situated in a wider social context. Therefore social phenomena should not be analyzed in isolation. A totalistic view indicates that all the components are interrelated into a coherent structure which can only make meaning in terms of the structure, but then the structure relies on the component parts.

Essence

This refers to the fundamental elements of analytic process. Critical social researchers see essence as a fundamental concept that can be used as the key to unlock the process of deconstruction.

Praxis

This refers to the practical reflective activity. It involves what you do most of the time as a human being. It excludes such instinctive or mindless activities like sleeping, breathing etc or activities that involve repetitive work tasks. It is what changes the world. The critical social researcher believes that knowledge not just about finding out things about the world. It is about changing it. You need therefore to engage in praxis.

Ideology

This is a concept which has a long history. Its current usage is developed as an analytic and critical tool in the work of Marx. It has been an important feature of Marxism. There are two approaches to a critical analysis of ideology. These are the positive and the negative views of ideology.

Structure

In critical social research, structure is viewed holistically as complex set of interrelated elements which are interdependent and which can only be adequately conceived in terms of the complete structure.

History

This refers to both the reconstructed account of past events and the process by which this reconstruction is made. That is the process of doing history. It involves both a view about the nature of history and the assembling of historical materials.

The Critical Research Process

The process here involves deconstruction and reconstruction. Note that this does no means taking a house apart brick by brick and may be building another house using that same bricks.

Reconstruction is not only rebuilding, it involves reconceptualization. Critical research starts with observation, concern, frustration or doubt which provoked the enquiry.

Approaches in Critical Social Research

There are four approaches in critical social research

Critical case study: Here you select for detailed empirical which provides a specific focus on analysis of a myth or contradiction. A variety of different data collection techniques can be used within a critical case study approach.

Radical Historicism: This presupposes that constructing histories is an interpretative process rather than the recording of facts it attempts to dig beneath the surface of the historical development of structural forms. Radical historicism is concerned with the uncovering of historical evidence.

The meaning of the evidence depends on a conceptualization of dominant social structures. The reconstruction of history takes place alongside structural analysis. It informs and is also informed by it.

Critical Ethnography: This is widely used and it involves a close attention to details which characterize ethnography. It is very useful in rendering all the invisible to be visible and for revealing anomalies and common sense notions. It transforms all the anomalies and all details taken for granted into contradictions and myths by situating them in broader social and historical analysis. It focuses on the way contradictions are negotiated and myths represented.

Structuralist techniques: The two types of structuralist techniques used in critical social research are:

- Semiological Analysis: This attempt to undercover the connoted level of denoted messages though widely used in the mass media. It is applicable to and derives from a general approach to the analysis of any system; it sees a sign as any cultural symbol which conveys a meaning. The sign is made up of two elements – signifier and signified.

- Identification of binary oppositions and narrative sequences: Which draws on linguistics and presupposes that the structure of language is inherently dichotomous and consequently, the symbolic meaning of an image is determined only by differences.

UNIT 8.0 PHILOSOPHICAL AND HISTORICAL RESEARCH

PHILOSOPHICAL RESEARCH

Philosophical studies aim at the assessment of the status of knowledge through analysis of meaning and relationships of different concepts and exposition of underlying assumptions.

They also aim at a fruitful synthesis of ideas from different fields concerning the theories and practices.

The main thrust of philosophical research can be summarized as follows:

Study of the contributions made by a philosophers or groups of philosophers: These are philosophers or groups of philosophers who follow similar lines of thought expressing their ideas about the aspects of occasions in different forms – speeches, discourses, writing and institutional practices.

Study of the philosophical ideas propounded by a particular school of thought: You already know that philosophy is classified under different schools of thought, such as Idealism, naturalism, realism, existentialism, pragmatism, socialism or communism i.e. dialectical materialism.

Study of the philosophical bases of the curriculum: So many questions are raised in the context of curriculum development. These include the desirability of the state of mind to learn, the criteria for identifying the desirability of the state of the mind.

Study of the philosophical bases of instructional process: Philosophical research probes in to the alternative to instructional designs and system, their worth and social desirability in the overall frame work and understanding to human development.

Study and philosophical analysis of contributions made by theories in psychology: Philosophical studies probe into the contributions of need theories and concept formation from the psychological process.

Study and philosophical analysis of social theories: Issues like freedom, autonomy, democratic values, equality of opportunities, policies and practices, require philosophical analysis from time to time in order to accommodate the changes which improve or develop the societies.

The Main Steps In Philosophical Inquiry

Identification of appropriate topic: As it is in every other research method, philosophical studies start with the selection of appropriate topic. You have to ensure that the topic chosen is capable of yielding a system, of thought with sound justification. Of course, you have to consult available literature before you finalize the topic. Look at these two topics.

Collection of data: After the identification of the topic and raising some preliminary questions about this topic, you then go to collect all possible data relevant to the topic. Data can be collected from different sources including available literature. The type and sources of data may be of literary nature like to rite ups or opinions of the philosophers concerned, commentaries on relevant philosophical works which appear in the form of books, journals, transcriptions, recording, research reports etc.

Classification of data and the interpretation: Logical classification of the data under different heads that focus on the topic of investigation is required here. After the classification comes the interpretation. This follows different processes such as description, comparison, appraisal, cross examination, etc. of the different ideas or concepts in the context of the major question under consideration, interpretation.

According to varma (1965) the task of interpretation is chiefly that of ascribing a significance, meaning, purpose and relatedness to a common end and to an apparently heterogeneous mass of data. Personal biases and subjectivity must be as much as possible avoided.

Reporting the study: This is the final stage of the research. You have to maintain logical sequences among the different heads of classification appropriate conclusion are drawn towards the end of the presentation.

HISTORICAL RESEARCH

Right from your primary school, you have known that history is the study of the past. Therefore a historical research consists of the studies of the past happenings. In this process of searching for the amount of what happened in the past, the historian may reveal several interacting factors that contributed to a particular event within a specific context of time and space occurring.

Historical studies highlight some causal relationships of past events and unearth the background for the purpose of understanding the social phenomena both past and the present.

History is a study of the road for the march of progress indicating the nature of advance in socio-economic and political contexts. According to Borg (1963) historical research can be seen as the systematic and objective location, evaluation and synthesis of evidence in order to establish facts and draw conclusions about past events.

The Main Features of Historical Research

- It aims at developing knowledge of past events within a particular framework of time and social – political, economic and cultural context.
- You dig into the significant data that tell us about the past events.
- Historical research is conducted based on the analysis of all the known information related to the research problem.
- Data concerning the past events are available through different sources.

These sources are classified into primary and secondary sources:

(A) Primary sources of Historical data

These provide first hand information about the past events. They include direct observation and reporting a recording of experiences. Other primary sources include:

- Personal primary sources such as personal direct observation of events which the participated in the past.
- Physical artifacts like collections in museum or evidence of historical spots, relics, remains etc.
- Mechanical artifacts like films, video, audio and photographs
- Records written by actual participants or observers in the form autobiographies, charters, court decisions, official minute or record deeds, wills, permits, licenses, certificates, bills, receipts, maps, pictures, paintings etc.

(B) Secondary Sources of data

These involve second-hand information about past events. The person giving the information is neither a participant nor an eye witness of events. The information can be in the form of written materials such as newspaper articles, magazine, books, research reports etc. You have to note that secondary sources of data are usually of limited worth because of the errors which result when information is passed on from one person to another.

The Main Steps In Historical Research

- Identification of the problem
- Specification of the population of data
- Data Collection
- Organization of Data
- Interpretation of data
- Report writing

UNIT 9.0 NATURALISTIC INQUIRY AND CASE STUDY

THE MEANING OF NATURALISTIC INQUIRY

As the name implies, this can be said to mean all investigations about social and educational phenomena conducted in natural settings. It is identified with the concept of field studies in the areas of Anthropology and Sociology. It has a unique position in the conduct of social sciences researches.

For instance, if you are interested in studying the problems of hotel management in Nigeria, you may decide to use the naturalistic inquiry method. In this case, it means that you have to go down to the natural settings of the hotels and their management in order to study the problems.

In whatever you are doing, objectivity must be ensured through the use of representative samples, standardized instruments and different sophisticated statistical designs for treatment of data and generalization of finds. Naturalistic inquiry does not follow these steps. It follows an altogether different conceptual framework this takes into account such factors as:

Multiple Realities: Naturalists believe that multiple realities exist in social situations. These exist in concrete forms. Realities can be regarded as what people perceive at a particular point in time.

Meanings and Interpretations: Naturalists lay emphasis on the meaning and interpretations given to objects, events and processes concerning social and educational situations.

Generation of Knowledge: Naturalists insist on the generation of knowledge as a result of the interaction between the researcher and the respondents. The respondents answer questions from the investigator in order to show their perceptions or the meanings attached to their actions.

Generalization: Naturalists do not believe in generalization as is done by the scientists. They believe that the process of knowledge generalization must take into account the differences or the real evidence existing in specific situations.

Human Relations: There are several intrinsic factors, events and processes that keep influencing each other. If is not possible to identify one to one cause and effect relationship in the case of naturalistic studies.

Value Systems: There is no value – free inquiry. Naturalists assume the influence of value systems in the identification of problems, selection of samples, use of tools for data collection, the conditions in which data are gathered, and the possible interaction that takes place between the investigator and the respondents.

The Uniqueness of the Naturalistic Inquiry

Holistic Approach: The intention of naturalists of the development of deeper understanding of a given situations in a holistic manner. All possible information in regards to all the significant dimensions of the situation under study is collected with a view to show the situation in the totality.

Insightful Inquiry: This is emphasized by the naturalists where human beings are treated as the sole means of data collection.

No a Prior Theory: A naturalistic research is such that you go to the field collect data without having any pre-specified theory in mind. Your belief as a naturalist is that and a prior limits the inquiry to those elements which may have been significant before developing an understanding of the situation.

No Pre-specified design of study: There is no explicit statement on the hypotheses and the conditions for data collection, analyses and interpretation.

Naturalistic setting: In the description of naturalistic research above, you learnt that it takes place in the natural setting. So naturalistic believe in conducting their studies in realistic settings since reality cannot be studied in fragmented and controlled situations.

The Main Steps In Naturalistic Inquiry

Identifications of broader questions of Inquiry: You have to first specify the pertinent issues or questions related to the settings and which can be resolved or answered through field study. Your main focus should be on the specific structure of occurrences rather than general character of the Phenomenon.

Collection of the initial level data: The next step is to make deliberate attempts to identify a full range of variations in the social and organisational arrangement related to the problem under study. You may start the investigation from a broader context of the problem before you proceed to the specific occurrences of the events.

Procedures for data collection: You can collect data in different phases through participant observation.

Devices of data collection: Different types of devices can be employed for the purpose of data collection. These include: writing notes about an observed situation; using electronic appliances such as tape recorders and video camera; taking photographs; and collecting relevant documents and literature on the problem.

Data Analysis: Data are analyzed by describing them. The frequency data are presented in two or three way contingency tables to show the patterns of behaviour. You can some of the times use descriptive or nonparametric tests like chi-square, Man Whitney, rank order correlation etc to identify certain patterns of relationships in the context of the specific situation under study.

A sound naturalistic research follows a cyclical process. This involves, data collection, generation of hypothesis, data examination, further generation and or modification of hypotheses, further data collection and verification until specific research questions are identified and the patterns of refined relations are arrived at.

Case Study Research

This can be defined as our intensive investigation about pertinent aspects of a particular unit in a given situation. This unit of investigation can be an individual, a family, a group of individuals or families, educational or social institutions, a community or a culture, a village, a tribe, a slum area, or even a hotel unit, a bank or group of banks or hotel etc.

A case study conducted on individuals may be linked with the processes of growth and development of a child, the behaviour of gifted children, psycho-analysis of a problem child, role of a leader in a specific social movement, the role of a manager in a company, etc.

Case studies are conducted for developing deeper understanding of intricate relationships existing in the process – aspects of a specific unit or unit through qualitative investigation. It is not very different from naturalistic methods. This is why most of the times; it is treated as kind of naturalistic research.

The Characteristics of Case Study List the Main Steps in Case Study

Continuity in investigation: A continuous and prolonged enquiry about the situations is very necessary till the underlying factors are explored and plausible patterns of their interaction or relationship are identified.

Completeness: A sound case study must involve extensive collection of data from internal and external environment of the unit under study. Data collection will continue till the completeness of data is ensured and a complete picture emerges.

Authenticity of Data: A case study report must be based on meaningful, reliable and valid information about the case. Appropriate applications of both qualitative and quantitative methods like interviews, observations, records, surveys and the administration of tests and questionnaires can be made.

Confidential recording: All data involving personal and ethnical issues like relationships of teachers and pupils with the management, discipline, confidential records, documents about

the institutions etc, must be handled tactfully and care must be taken to maintain their secrecy.

Intellectual synthesis: You know that case study involves multi-method of inquiry and deals with all significant situations concerning the unit, appropriate synthesis of the data is very necessary in order to show the uniqueness of the unit and to explore significant relationships.

The Main Steps In Case Study

Selection of a case for investigation: In selecting your case for investigation, there are some basic questions:

Are you interested in the study of a normal situation with a view to developing deeper in the phenomenon?

Are you determined to solve the problems of a typical institution?

Are you assigned the job of evaluating the functioning of an institution?

Are you interested in identifying the underlying factors contributing to the excellent performance of an institution?

Data Collection: The process of data collection can be through both qualitative and quantitative techniques like observations, interviews, check lists, Performa, open ended questionnaires, surveys, records, psychological tests etc. Personal interaction should come first. Then care must be taken to make sure that only tools relevant to the case are used.

Analysis of first round data: This is done systematically in order to identify the more complicated situations or problems and therefore raise important questions about the influential factors.

Second Round Investigation: This is conducted for these specific questions or factors which are identified during the first round data analysis. This second round involves intensive investigation through prolonged observations. Formal and informal interviews, questionnaires, cross-examination of different documents and record, administration of specific test etc. After this round analysis and interpretations of data begin.

Introduction of alternative measures: This is mostly done in clinical studies where most of the suitable alternatives as hypothesized through investigations are introduced.

Follow-up activities: The effectiveness of the alternative measures introduced should be investigated. This will give a feedback on the strengths and weaknesses of the corrective measures.

UNIT 10.0 DESCRIPTIVE RESEARCH

THE TYPES OF DESCRIPTIVE RESEARCH

Survey Studies

Most of the times, descriptive research is called survey research. But it is better to call survey a category under descriptive research survey itself is the most widely used method for obtaining descriptive and evaluative information. This can be used when trying to solve problems in education, government, industry, politics, organization etc.

Data may be gathered from every member of a specified population or from a carefully selected sample. The steps involved in survey research are in general, similar to what obtains for descriptive research:

Planning: This involves determining what topic to be studied, the type of population to be investigated and the methods and procedures to be used for data collection.

Sampling: This involves a decision about which people from the population to be included in the survey. Note that if you are going to generalize from the sample to the population.

Development of data collection instrument: This involves listing the questions and planning for the format of the instrument to be used. Is it personal interview, questionnaires, rating scales etc?

Carrying out the survey: This includes pre-testing the instruments to see if it will yield the desired data, interviewing the subjects or administering the questionnaires and verifying the accuracy of the data collected.

Processing the data: This includes tabulating the data, analyzing the data (you can use computers if you like) interpreting the results and reporting the findings.

Analysis of Documents

Documents and records can be used to bring out pertinent data. Documentary analysis otherwise referred to as context activity or informational analysis is very much like the historical research. But while historical research is primarily concerned with the more distant past, descriptive research is concerned chiefly with the present. There are wide varieties of documentary surveys made; some researchers can analyze judicial decisions, state laws or court rulings.

Also newspaper, periodicals, motion pictures, cartoons and other sources can also help. Some of the advantages include:

- To describe the prevailing practices or conditions in the field
- To discover the relative importance of, or interest in certain problems or topics i.e. to spot trends.
- To discover the level of difficulty of presentations available in text books or inother publications.
- To evaluate bias, prejudices or propaganda in text book presentations
- To analyze the types of errors and weakness in performance.
- To evaluate the relationships of stated objectives and what is being done practically.
- To identify the literary style, concepts or beliefs of a writer.

Correlational Studies

The value of Correlational research is that it is able to achieve this end. You know that one of the primary purposes of science is to discover relationships among phenomena with a view ultimately to predicting and, in some situations, controlling their occurrences. Most of the researches in social sciences and education are concerned with the establishment of interrelationships among variables.

Correlational studies are concerned with determining the extent of relationship existing between variables they enable us to measure the extent to which variations in one variable are associated with variations in another.

Relational Studies: Which is particularly useful in exploratory studies in the fields where title or no previous research has been conducted. It is a shot in the dark in order to verify haunches which you have about presumed relationships between some characteristics of variables.

Prediction Studies: Which are usually conducted in an area having a firm and secure knowledge base? It is based on the assumption that at least some of the factors that will lead to the behaviour to be predicted are present and measurable at the time the prediction is made.

Causal-Comparative Studies

Causal comparative studies provide the means for tackling problems that cannot be probed in laboratory situations. Again they yield valuable information and elites about the nature of phenomena and are well suited to many types of field studies seeking to establish causal relationships. Their limitations are:

- Lack of control is a serious limitations and weakness
- It is usually difficult to identify the relevant factors causing the particular conditions or phenomena
- When a relationship between variable is established, it is difficult to distinguish between the cause and the effect.
- The classification of subjects into dichotomous groups for the purpose of comparison also presents problems
- In comparative studies of natural situations, you do not have the same control over the selection of subjects as you have in experimental studies.

UNIT 11.0 EXPERIMENTAL AND ACTION RESEARCHES

The Experimental Research

The studies here are designed from establishing causal relationships. The method begins with a question concerning the relationship between two or more variables. In order to do this, you propose one or more hypotheses stating the nature of the expected relationship. Thus an experiment is the events planned and carried out so as to enable you collect evidence relevant to the hypotheses.

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Kerlinger (1974) considers experimental research the idea of science because answers to research questions obtained in experiments are on the whole clearer and less ambiguous than answers obtained in non-experimental research types.

An experimental research may be described as a research study in which one are more independent variables are manipulated and in which all or nearly all possible influential variables not pertinent to the problem of investigation kept to a minimum.

An experimental research in its simplest form has three characteristics. These are

- The manipulation of the independent variable.
- All other variables except the independent variable are hold constant
- The effect of the manipulation of the independent variable on the dependent variable is observed.

The Characteristics of Experimental Research

In the conduct of an experiment, there are three major ingredients. These are Control, Manipulation and Observation

Control: This is very important in experimental studies because without it, it is impossible to evaluate unambiguously the effect of an independent variable.

Manipulation: This refers to a deliberate operation performed by the researcher. In the behavioural sciences and education, the manipulation of a variable takes a characteristic form in which the experimenter imposes a predetermined set of varied conditions on the subjects. This set of varied conditions is referred to as the independent variable, the experimental variable or the treatment variable.

Observation: In an experimental study, we are interested to know the effect of the manipulation of the independent variable on a dependent variable. Therefore you make observations with respect to some characteristics of the behaviour of the subjects used in the research. These observations, which are most of the times quantitative in nature, may constitute the dependent variable.

The Steps in Experimental Design

There are number of steps involved in experimental studies. The steps that will lead you to the actual experiment are:

- Making a survey of the literature related to the problem.
- Identification and definition of the problem
- Formulating the hypotheses and defining basic terms and variables.
- Construction of the experimental plan.
- Identification of all the non-experimental variables that might contaminate the experiment and determine how to control them
- Selection of a research design
- Selection of the sample of subjects to represent a given population and the assignment of the subjects to groups and experimental treatments to the groups.

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- Construction or selection and validation of the instruments for measuring the outcomes of the experiment.
- Outlining the procedures for collecting the data and possibly conducting a plot or trial test to perfect the instrument or design.
- State the statistical or null hypothesis.

The Types of Experimental Designs

These are so classified because they offer minimal control of extraneous variables. Let us see examples.

One group Pre-test, Post-test design

This involves three steps:

- Administering a pre-test measuring the dependent variable
- Applying the experimental treatment X to the subjects
- Administering a post-test measuring the dependent variable.

Static Group Comparison

This design uses two or more groups. Only one of them is exposed to experimental treatment X. However the groups are assumed to be equivalent in all relevant aspects except in treatment.

TRUE EXPERIMENTAL DESIGN

This is so called because it provides:

- Random assignment of subject to the groups
- Random assignment of treatment to the groups
- Post testing of all groups.

Description of Action Research

According to Cohen and Marion (1989) action research is a small – scale intervention in the functioning of the real world and a close examination of the effects of such intervention. It is characterized by:

- Situational: It is derived out of situational needs and a solution is also designed with regards to the situation.
- Collaborative and Participation: It can be carried out on individual basis. But it is increasingly becoming a team work where practitioners and colleagues in an organization collaborate and participate with the researchers.

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- Self evaluation: This is necessary where the research team evaluates the outcome of the exercise. Remember that the study is self-initiated and evolves out of the perception of the problems by the practicing individual or group.

THE STAGES OF ACTION RESEARCH

There are four main stages in the conduct of action research:

Stage One: Diagnosis

At this stage we have the following:

- Identification, evaluation and formulation of problems i.e. diagnosing the problem on the basis of the symptoms.
- Preliminary discussions and negotiations among interested parties
- Review of literature – although not in all cases, but some types of problems may call for this before articulating the problem.
- Modification and redefinition of initial statement of problem.

Stage Two: Planning and Intervention

This involves the selection of the research procedure-designing the intervention, sampling, administration, choice of material and methods of evaluation.

Stage Three: Intervention and impact assessment

This is the actual implementation, collection and analysis of data to assess the change in the magnitude of the problem. It involves:

Implementation of the project: That is actually carrying out the treatment designed on the sample. It will include collection of relevant data.

Interpretation of data: Using minimal statistical or qualitative analysis, the data collected can be interpreted so that the impact of the treatment on the alleviation of the problem can be assessed.

Stage Four: Reflection

At this stage you look beyond the factual data based result, with the main purpose of interpreting the results. Reflect on the why‟s and how‟s of the finding and the onward destination.

UNIT 12.0 POPULATION AND SAMPLING

POPULATION

A population is any group of individuals that have one or more characteristics in common that are of interest to the researcher. Population parameters and sampling techniques are of

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paramount importance and are regarded as very critical factors for the success of most researches.

The primary purpose of research is to discover principles that have universal application. To study the population in order to arrive at generalizations would be impracticable, if not impossible. Some populations are so large that their characteristics cannot be measured. Before the measurement may be completed, the population itself would have changed.

DESCRIPTION OF A SAMPLE

A sample is a small proportion of a population selected for observation and analysis. The process of sampling makes it possible to draw valid inference or generalizations on the basis of careful observation of variables within a relatively small proportion of the population.

Using the characteristics of the sample, you can make certain inferences about the characteristics of the population from which it is drawn. Samples are not selected haphazardly. They are selected in a systematically random way, so that chance or probability operations can be used. Remember, a sample consists of elements selected from a population.

Failure to define the population clearly can make generalization from the sample observation highly ambiguous and result in drawing in accurate conclusions. The sample should be a representative number of respondents from the defined population.

THE PURPOSE OF SAMPLING

The major purpose of studying samples instead of the entire population is that the population is sometimes too large that it is not feasible to study it. So sampling allows you to study a workable number of cases from a large group to get findings that are relevant to all members of the group.

TYPES AND PROCEDURES FOR SAMPLING (SAMPLING TECHNIQUES)

Sampling techniques are simply defined as methods of drawing samples from total populations. They are classified into two: Probability and Non-probability sampling techniques

Probability Sampling

In this technique, the components of the samples are chosen from the larger population by a process known as randomization. The principle of randomization ensures that all the individual subjects of the larger population have equal chances of being selected and included to be members of the drawn samples. This generates valid samples that are truly representative of the larger populations.

Non-Probability Sampling

This is a type of sampling where you do not know the probability of each population element being included in the sample. They are known to generate biased samples that are not truly

representative of the target population from which they are drawn. Any sample that is non-valid can encourage errors in findings that cannot be generalized to the total population.

The greatest limitations of the non-probability samples are:

- Because there is no probability in the selection of the elements for the samples no real claim of representativeness can be made.
- The degree of sampling errors remains unknown and unknowable.
- All the statistical test of significance is based on laws of probability and assume that a random process is used in selecting the samples.

Differentiate between probability and non-probability samples?

SAMPLING TECHNIQUES

In the section above, you learnt that the two broad sampling techniques are probability and non-probability sampling. Each of these sampling techniques has a number of sampling approaches or methods under it. For instance the probability sampling has under it, simple random, systematic, stratified and cluster sampling.

- **Simple random sampling**

This is based on the principle that individuals are chosen from a target population such that each has an equal chance of being selected, and each choice is independent of the other.

- **Systematic Sampling**

If your target population can be accurately listed, then this method can be used to get what approximates a random sample. It consists of the selection of a Kth number from the list. For instance, if you have 100 customers and you want select 10 out of the 100 as your sample. You will divide 100 by 10 to get 10. From a list containing 1-10 select any number at random. Let us say 8. In this method all the numbers connected to 8 are automatically selected. So you will have 8, 18, 28, 38, 48, 58, 68, 78, 88, and 98, as your samples.

- **Stratified Random Sampling**

In the case the population is sub-divided into smaller subgroups called strata, e.g. male, female, urban, rural, Christians, Muslim, traditionalist etc. Then simple random sampling is employed to draw required samples from each of the strata.

- **Cluster sampling**

When the population is spread across a large area, it becomes difficult to make a list to select samples for study through the normal simple random process. The large area has to be subdivided into sub-units for effective handling. For instance a state could be sub divided into zones, local governments or wards and a series of cluster sampling is done from each of these sub units to get the sample which will be truly representative of the target population.

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Non – Probability Sampling

This type of method uses whatever subjects are available rather than following a specific subject selection process. This method produces samples which do not accurately reflect the characteristics of the population of interest. Such samples may lead to unwarranted generalizations and should not be used if randomization is possible.

Under this we may have:

1. **Accident or Continuance Sampling**: This takes the subjects as they come to the scene or as they are presented to you by mere happenstance. There is no attempt to control biases.

2. **Quota Sampling**: This selects the respondents in the same ratio as they appear or are found in the target population. It is also an unregulated type of sampling. This is because you have no probability of how the subjects will be selected. Others in this category are Availability Sampling, Purposive Sampling, and Dimensional Sampling etc.

SAMPLING SIZE

According to Best and Khan (1995) the ideal sample is large enough to serve as an adequate representative of the population about which the researcher wishes to generalize and small enough to be selected economically in terms of subject availability, expenses in both time and money, the complexity of data analysis etc.

In other-words there is no fixed number or percentage of subjects that determines the size of an adequate sample. The size may depend on the nature of the population of interest or the data to be collected and analyzed. Samples of 30 and above are considered large samples and less than 30 are considered small samples.

However, it is important that care must be taken in the selection of the samples. Emphasis should be on representativeness rather than sample size. Random sampling or letting chance or law of probability determine which members of the target population are to be selected is the ideal thing to do

UNIT 13.0 STATISTICAL TOOLS OF ANALYSIS

DATA ANALYSIS: DEFINITION AND MEANING

Nwana (1981) has described data analysis as referring to those techniques whereby the investigator extracts from data, information that was not apparently there before and which would enable a summary description of the subject being studied, whether human or material, to be made. To

Kerlinger (1971) data analysis is the categorizing, ordering, manipulating and summarizing of data, adding that its purpose is to reduce large quantities of raw data to manageable and

interpretable form so that characteristics of situations, events and people can be succulently described and the relations among variable studied and interpreted.

WHAT IS STATISTICS?

Statistics is a body of mathematical technique or processes for gathering, organizing analyzing and interpreting numerical data. You know that most research yields such quantitative data. Therefore statistics is a basic tool of measurement, evaluation and research. The concept statistics is sometimes used to describe the numerical data that are collected. Statistical data describe group behaviour or group characteristics abstracted from a number of individual observations that are combined to make generalizations possible.

According to Kerlinger (1971) statistics is considered the theory and method of analyzing data obtained from samples of observations in order to describe population, to study and compare sources of variances, to help make decisions to accept or reject relations among phenomena, and to aid the process of making reliable relations among empirical observations.

Measurement is the most precise and universally accepted process of description, assigning quantitative values to the properties of objects and events.

PARAMETRIC AND NON-PARAMETRIC STATISTICS

For parametric statistics, we assume that most populations have at least one parameter. A parameter is a characteristic or quality of a population that, in concept is a constant but whose value is a variable. In statistics parameters refers to characteristics of a population. But when you have similar characteristics of a sample, these sample characteristics can be called statistic.

Non Parametric Data: These are data which are either counted or ranked. They are sometimes known as distribution free tests. They do not depend on the assumption of normally distributed populations.

Descriptive Analysis: In this type of statistical analysis, generalizations are limited to the particular group of individuals observed. The conclusions are not extended beyond the group.

Inferential Analysis: This type of statistical analysis involves the process of sampling and the selection of a small group that is assumed to be representative of the population from which it is drawn. So drawing conclusions about population based on observation of the samples is the purpose of inferential analysis.

MEASURES OF CENTRAL TENDENCY OR LOCATION

These are used to indicate the central point around which data revolve. They are of several forms. The most common is statistical analysis of data are the mean, median and mode. They help you to find out from your data of numerical values what values can be representative of the sample of people or objects which you are studying.

Mean

Mean is the arithmetic average of the values in a distribution. It is obtained by adding up all the values in the distribution or group of scores and dividing by the total number o values. It is the most useful of all statistical measures, because, in addition to the information that it provides, it is the base from which many other important measures are computed.

For instance if

$$X = 2, 3, 4, 5, 6, 7, 8$$

$$\text{Then } \sum X = 2+3+4+5+6+7+8 = 35$$

$$N = 7$$

$$X = {}^{35}/_{7} = 5$$

The Median

This is a point or value that divides the distribution of values into two equal groups with as many cases above it as below it. It is a measure of position rather than of magnitude. It can be found by both inspection and calculation. When we have single values we use inspection. But if the values are grouped we use calculation.

For instance if

It can be found by both inspection and calculation. When we have single values we use inspection. But if the values are grouped we use calculation. For instance if

X = 2, 3, 4, 5, 6, 7, 8. You can see that the middle number is 5. It is the median. But if we have see 2, 3, 4, 5, 6, 7, 8, 9. The middle number is added and divided by 2. In this distribution 5 + 6 = 11 ÷ 2 = 5.5. The median is **5.5**

Mode

This is the value which is characterized as the highest number of frequencies in a frequency distribution of values. It is the value that occurs most frequently in a distribution. It can be located by inspection when values are listed or by calculation when values are grouped.

For instance:

2, 3, 3, 4, 4, 4, 5, 6, 6, 7, 8, 9. You can see that the highest occurring number in this set of number is 4. It is the mode.

MEASURES OF VARIABILITY OR DISPERSION

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These tell us how the values spread in a scale of distribution. According to Nwana (1981) the variability of a set of values can contribute immensely to the need for researchers to determine objectivity, the extent to which marks, weights, heights, views, opinions and infact all values of a variable obtained in a research do differ from one another.

Let us look at some of these measures.

The Range

Range is the difference between the highest value and the lowest value in a set of data. It is the difference between the two extreme values in a distribution of figures. It is an unreliable measure, because it does not guarantee the researcher a fair and dependable index of variability.

This is because of the use of the two extreme values. If you look at the mathematics test in activity 1, you will see the highest number is 95 and the lowest number is 55.

Therefore the range is 95 – 55 = 40.

Mean Deviation

This is the average distance away from the mean in which the measure lies. It is used by the researcher to determine the power of representativeness of the mean as a measure standing in for other measures in the same distribution. A value expressed as its distance from the mean is called a deviation value. It is given by the formula X = (X – X). For the mean deviation it will be X. (X – X) 2 /N. This can be done regardless of the signs.

Variance (δ 2)

The value of the sum of the deviations from the mean is always zero. So when you square the deviations around the mean and sum up the value and divide by N what you get is the variance. It will give you a positive value. It is given by $\delta 2 = \sum(X - X) 2$. The variance is a value that describe how all the values in a distribution are dispersed to spread about the mean. It is very useful in describing the characteristics of a distribution and will be employed in a number of very important statistical tests.

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X	(X – X)	$(X - X)^2$	Variance $\delta^2 = \frac{\sum(X - X)^2}{N}$
70	-5	-25	
65	-10	100	
60	-15	225	
55	-20	400	
75	0	0	
80	+5	25	
85	+10	100	
90	+15	225	
95	+20	400	

$$\sum(X - X)^2 = 1500$$

Standard Deviation

This is the square root of the variance. It is the most frequently used measure of variability or spread in a distribution. According to Cohen (1976) it is the most reliable of all the available measures of dispersion. It is given by

$$\delta^2 = \sqrt{\frac{\sum(X - X)^2}{N}} \text{ or } \sqrt{\frac{\sum X^2}{N}}$$

In the above example where variance $\delta^2 = 1666$.

67 The standard deviation will be 166.67 = 12.91

Both the variance and standard deviation can be calculated using the raw score method. It is given by

$$R^2 = \frac{N\sum X - (\sum X)^2}{N2} \quad \delta = \frac{N\sum X - (\sum X)^2}{N2}$$

The Normal Curve

The law of probability and the normal curve that illustrate if it is based on the law of chance or the probability occurrence of certain events. If a set of observations conforms to the mathematical form it is represented by a bell-shaped curve which has definite characteristics.

Features of a Normal Curve

- The curve is symmetrical around its vertical curve axis.
- The terms cluster the centre or median
- The mean, median and the mode have the same value
- The curve has no boundaries in either direction for the curve
- The cover touches the base line no matter how far it is extended.
- The curve is a curve of probability not certainty.

The Characteristics of the Normal Curve

- It is symmetrical – the percentage of frequencies is the same for equal intervals below or above the mean.
- The terms of values cluster or crowd around the mean
- The terms curve is highest at the mean.
- The curve has no boundaries.

Applications of the Normal Curve

- To calculate the percentile rank in a normal distribution
- To normalize a frequency distribution an important process in standardizing a psychological test or inventory.
- To test the significance of observed measures in an experiment.

MEASURES OF ASSOCIATION OR RELATIONSHIP

These measures called correlation coefficients are concerned with the determination of the relationships between variables and variables. They are mostly used in determining whether or not there is a relationship is always expressed in some degree of coefficient the degree of relationship, direction, and amount are always represented by the index of positive or negative called correlation is a method in which an index of coefficient.

Technically, in quantitative research, correlation is a method in which an index of coefficient is calculated to describe the degree of relationship or association between two sets of paired values and then tested to determine the probability or the chances of occurrence of the relationship. Correlation is indicated by +1.00 No correlation or association is indicated by 0.00. While perfect negative relationship is indicated by -1.00.

PARAMETRIC TESTS

These are the powerful tests in statistics and should be prefentially used if their base assumptions are not. The assumptions are related to the nature of the population distribution and on the way the type of scale is used to quantify the data observations.

Decision Making

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Statisticians do not deal with decisions based on certainty. They estimate the probability or otherwise of occurrence of events when you reject a null hypothesis when it really true and should of have been rejected, there is an error. This called type I error when you accept a null hypothesis when it is really false and should have been rejected, there is also an error, and this called type II errors. So you set the level of significance you will consider the relative seriousness of making type I or type II error.

Non Parametric tests

1. The Chi-square ($\lambda 2$): This is employed in comparism between observed and theoretical or expected frequencies or in testing the mathematical fit or frequency of an observed frequency distribution. So for frequency evaluation in certain research instances this is the most appropriate statistical method.

2. The Mann Whitney U Test: This is the counterpart of the t-test in parametric measurements. It is used for a two-condition unrelated design when different subjects are used for each of the conditions.

3. Wilcoxon signed ranked tests: This is used for a two-condition related design when the same subjects perform under both conditions i.e. matched subjects.

4. Friedman tests: This extension of the Wilcoxon tests. It is used for a related design if the same subjects or matched subjects are performing under three or more conditions.

UNIT 14.0 WRITING RESEARCH REPORTS AND PROPOSALS

CHAPTER AND SECTIONS CONSTITUTING A RESEARCH PROJECT REPORT

Speaking technically, the research process is a model of the scientific method of thinking or of analyzing problems. As a model of science, the research process in its simplified form will be constituted by the following stages of activities:

- Problem identification and definition
- Hypothesis or answer(s) formulation
- Research design and data collection
- Data analysis and discussion
- Conclusion and generalization making

A particular thesis need not have all the sections therefore; relevant sections would be used in the appropriate order. The format is as follows:

Preliminary pages:

- Title page
- Acceptance page or approval page

- Dedication
- Acknowledgement
- Abstract
- Table of content
- List of tables
- List of figures
- List of appendices

Chapter One- Introduction:

Background to the problem or Rationale for the problem

Statement of the problem

Significance of the study

Objectives of the study or purpose of the study

Scope of the study

Area of study / context of the study

Research questions and or hypothesis

Definition of terms (operational definitions)

Review of literature:

Theoretical and conceptual frame work

Review of related researches

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Research Methodology

- Research design
- Population
- Sampling technique and samples
- Instrumentation – development and administration of instruments
- Data collection
- Data analysis techniques
- Limitations

Results and discussion

- Presentation and analysis of data
- Interpretation of the findings

Summary and Conclusion

- Summary of results.
- General conclusion
- Implications of the study and / or recommendations.
- Suggestion for further study

g.Supplementary

- Bibliography
- Appendix
- Index

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